

# **CS GY 6083 - PRINCIPLES OF DATABASE SYSTEMS**

## **Milestone 3: DB setup and SQL**

### **Pet Adoption System**

Team Members:

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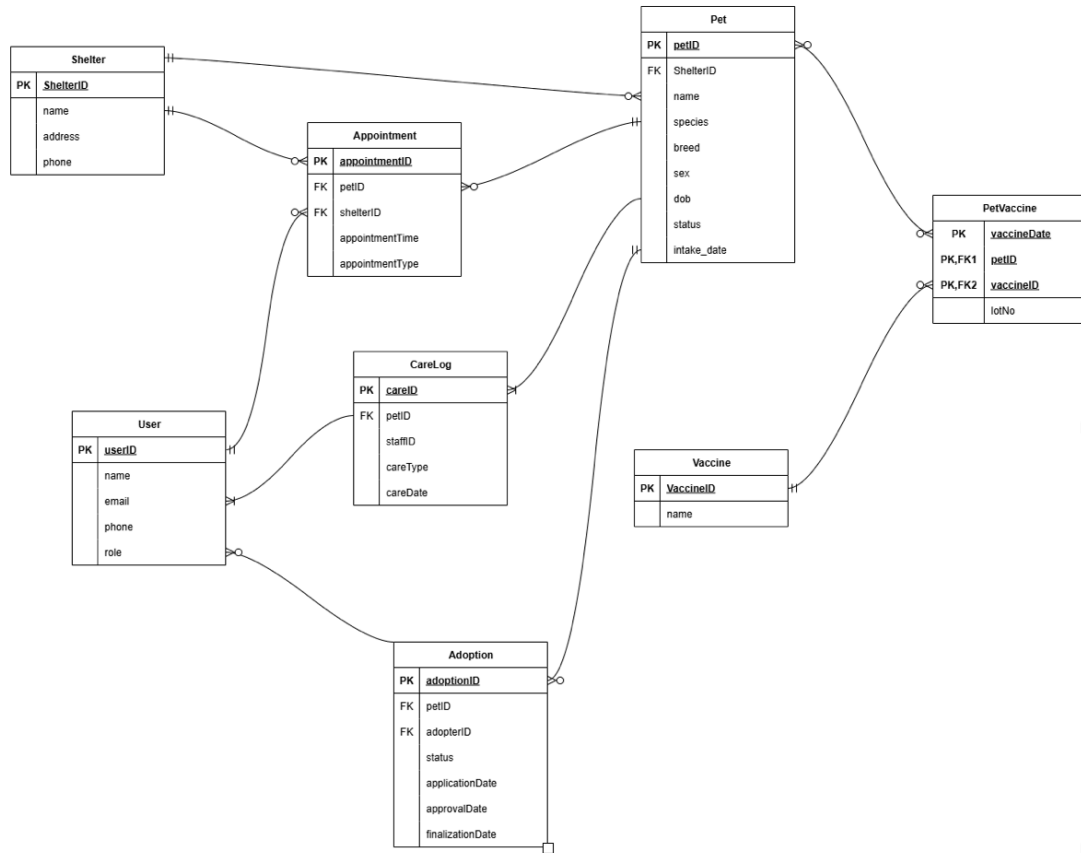
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Debika Dharma Lingam

#### **Purpose:**

- Set up the database to be used by your app
- Set all of your tables in your database
- Populate the database with the chosen dataset — be sure that the data is practical and sufficient
- Include finalised ER diagram
- Write all SQL commands you plan to use in your app
- Write at least 2 different PL/SQL commands (such as trigger, assertion, check constraint, stored procedure, Function,...) to efficiently manage the business logic of your app.
- Test and verify that your SQL commands work properly.

## Finalised ER Diagram and Schema:



### Schema

User (**userID**, name, email, phone, role)

Shelter(**shelterID**, name, address, phone)

Pet(**petID**, @shelterID, name, species, breed, sex, dob, status, intake\_date)

Adoption(**adoption\_id**, @petID, @adopterID (fk from user.userID(adopter), status, applicationDate, approvalDate, finalizationDate)

Appointment(**appointmentID**, @petID, @adopterID (fk from user.userID), @shelterID, appointmentTime, appointmentType)

Vaccine(**VaccineID**, name)

PetVaccine(@**petID**, @**vaccineID**, **vaccineDate**, lotNo)

CareLog(**careID**, @petID, @staffID (fk from user.userID(staff)), careDate, careType)

### UI components with tables matching:

1. Landing Page → Pet, Shelter tables
2. Login Page → UserAccount table
3. Shelter Owner's Dashboard → Pet, UserAccount, Adoption, Appointment, CareLog tables
4. Prospective Adopter's Home → Pet, Shelter, Adoption tables
5. Pet Details Page → Pet, PetVaccine, Vaccine, Breed tables + fn\_pet\_age\_years function
6. Appointment Booking → Appointment table + sp\_schedule\_appointment procedure
7. Adoption Application → Adoption table + trigger validation
8. User Account Dashboard → UserAccount, Adoption, Appointment, Payment tables
9. Staff Care Log → CareLog, Pet, UserAccount tables

## Database set up:

### 1) Database Server:

Our database is running locally using MySQL via phpMyAdmin/XAMPP as we are working within a Django framework to build our pet adoption system.

### 2) Table Schemas:

Below, we have attached screenshots of the table schemas for all our tables.

2 a)

The image displays two screenshots of the phpMyAdmin web interface. The top screenshot shows the results of a SQL query: `SELECT COUNT(*) FROM UserAccount;`. The result is a single row with the value 10. Below the result, there are buttons for 'Print', 'Copy to clipboard', 'Export', 'Display chart', and 'Create view'. A 'Bookmark this SQL query' section is also visible, with a label field and a checkbox for 'Let every user access this bookmark'. A warning message at the bottom states: 'Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.' The bottom screenshot shows the results of another SQL query: `SELECT COUNT(*) FROM Shelter;`. The result is also a single row with the value 10. Similar buttons and a 'Bookmark this SQL query' section are present. A warning message at the bottom of the second screenshot states: 'Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.'

Your SQL query has been executed successfully.

DESC UserAccount;

[\[ Edit inline \]](#) [\[ Edit \]](#) [\[ Create PHP code \]](#)

Extra options

| Field  | Type                                  | Null | Key | Default | Extra          |
|--------|---------------------------------------|------|-----|---------|----------------|
| userID | int(10) unsigned                      | NO   | PRI | NULL    | auto_increment |
| name   | varchar(100)                          | NO   |     | NULL    |                |
| email  | varchar(50)                           | NO   | UNI | NULL    |                |
| phone  | varchar(30)                           | YES  |     | NULL    |                |
| role   | enum('ADOPTER','STAFF','VET','ADMIN') | NO   |     | NULL    |                |

Query results operations



Print



Copy to clipboard



Create view



Bookmark this SQL query

Label:

☐ Let every user access this bookmark

[Bookmark this SQL query](#)

Your SQL query has been executed successfully.

DESC Shelter;

[\[ Edit inline \]](#) [\[ Edit \]](#) [\[ Create PHP code \]](#)

Extra options

| Field     | Type             | Null | Key | Default | Extra          |
|-----------|------------------|------|-----|---------|----------------|
| shelterID | int(10) unsigned | NO   | PRI | NULL    | auto_increment |
| name      | varchar(100)     | NO   |     | NULL    |                |
| address   | varchar(100)     | NO   |     | NULL    |                |
| phone     | varchar(30)      | YES  |     | NULL    |                |

Query results operations



Print



Copy to clipboard



Create view



Bookmark this SQL query

DESC Pet;

[ [Edit inline](#) ] [ [Edit](#) ] [ [Create PHP code](#) ]

Extra options

| Field      | Type                               | Null | Key | Default   | Extra          |
|------------|------------------------------------|------|-----|-----------|----------------|
| petID      | int(10) unsigned                   | NO   | PRI | NULL      | auto_increment |
| shelterID  | int(10) unsigned                   | NO   | MUL | NULL      |                |
| name       | varchar(100)                       | YES  |     | NULL      |                |
| species    | varchar(50)                        | YES  |     | NULL      |                |
| breed      | varchar(50)                        | YES  |     | NULL      |                |
| sex        | enum('M','F','UNKNOWN')            | NO   |     | UNKNOWN   |                |
| dob        | date                               | YES  |     | NULL      |                |
| status     | enum('AVAILABLE','HOLD','ADOPTED') | NO   | MUL | AVAILABLE |                |
| intakeDate | date                               | NO   |     | NULL      |                |

Query results operations



Print



Copy to clipboard



Create view



Bookmark this SQL query

Label:



Let every user access this bookmark

Bookmark this SQL query

Your SQL query has been executed successfully.

DESC Vaccine;

[ [Edit inline](#) ] [ [Edit](#) ] [ [Create PHP code](#) ]

Extra options

| Field     | Type             | Null | Key | Default | Extra          |
|-----------|------------------|------|-----|---------|----------------|
| vaccineID | int(10) unsigned | NO   | PRI | NULL    | auto_increment |
| name      | varchar(120)     | NO   | UNI | NULL    |                |

Query results operations



Print



Copy to clipboard



Create view

Console

mark this SQL query

Your SQL query has been executed successfully.

DESC PetVaccine;

[\[ Edit inline \]](#) [\[ Edit \]](#) [\[ Create PHP code \]](#)

Extra options

| Field       | Type             | Null | Key | Default | Extra |
|-------------|------------------|------|-----|---------|-------|
| petID       | int(10) unsigned | NO   | PRI | NULL    |       |
| vaccineID   | int(10) unsigned | NO   | PRI | NULL    |       |
| vaccineDate | date             | NO   | PRI | NULL    |       |
| lotNo       | varchar(64)      | NO   |     | NULL    |       |

Query results operations



Print



Copy to clipboard



Create view



Bookmark this SQL query

Label:

☐

Let every user access this bookmark

[Bookmark this SQL query](#)

Your SQL query has been executed successfully.

DESC Adoption;

[\[ Edit inline \]](#) [\[ Edit \]](#) [\[ Create PHP code \]](#)

Extra options

| Field            | Type                                                         | Null | Key | Default |
|------------------|--------------------------------------------------------------|------|-----|---------|
| adoptionID       | int(10) unsigned                                             | NO   | PRI | NULL    |
| petID            | int(10) unsigned                                             | NO   | MUL | NULL    |
| adopterID        | int(10) unsigned                                             | NO   | MUL | NULL    |
| status           | enum('APPLIED','APPROVED','FINALIZED','REJECTED','CANCELED') | NO   |     | NULL    |
| applicationDate  | datetime                                                     | YES  |     | NULL    |
| approvalDate     | datetime                                                     | YES  |     | NULL    |
| finalizationDate | datetime                                                     | YES  |     | NULL    |

Query results operations



Print



Copy to clipboard



Create view

Console

Your SQL query has been executed successfully.

DESC Appointment;

[\[ Edit inline \]](#) [\[ Edit \]](#) [\[ Create PHP code \]](#)

Extra options

| Field           | Type                             | Null | Key | Default | Extra          |
|-----------------|----------------------------------|------|-----|---------|----------------|
| appointmentID   | int(10) unsigned                 | NO   | PRI | NULL    | auto_increment |
| petID           | int(10) unsigned                 | NO   | MUL | NULL    |                |
| adopterID       | int(10) unsigned                 | NO   | MUL | NULL    |                |
| shelterID       | int(10) unsigned                 | NO   | MUL | NULL    |                |
| appointmentTime | datetime                         | NO   | MUL | NULL    |                |
| appointmentType | enum('VISIT','MEET&GREET','VET') | NO   |     | NULL    |                |

Query results operations



Print



Copy to clipboard



Create view



Bookmark this SQL query

Label:

☐

Let every user access this bookmark

[Bookmark this SQL query](#)

Your SQL query has been executed successfully.

DESC CareLog;

[\[ Edit inline \]](#) [\[ Edit \]](#) [\[ Create PHP code \]](#)

Extra options

| Field    | Type             | Null | Key | Default | Extra          |
|----------|------------------|------|-----|---------|----------------|
| careID   | int(10) unsigned | NO   | PRI | NULL    | auto_increment |
| petID    | int(10) unsigned | NO   | MUL | NULL    |                |
| staffID  | int(10) unsigned | NO   | MUL | NULL    |                |
| careDate | datetime         | NO   |     | NULL    |                |
| careType | varchar(80)      | NO   |     | NULL    |                |
| notes    | text             | YES  |     | NULL    |                |

Query results operations



Print



Copy to clipboard



Create view

Console

DESC Payment;

[\[ Edit inline \]](#)
[\[ Edit \]](#)
[\[ Create PHP code \]](#)

Extra options

| Field         | Type                                   | Null | Key | Default | Extra          |
|---------------|----------------------------------------|------|-----|---------|----------------|
| paymentID     | int(10) unsigned                       | NO   | PRI | NULL    | auto_increment |
| adopterID     | int(10) unsigned                       | NO   | MUL | NULL    |                |
| adoptionID    | int(10) unsigned                       | YES  | MUL | NULL    |                |
| appointmentID | int(10) unsigned                       | YES  | MUL | NULL    |                |
| amount        | decimal(10,2)                          | NO   |     | NULL    |                |
| paymentDate   | datetime                               | NO   |     | NULL    |                |
| paymentMethod | enum('CASH','CARD')                    | NO   |     | NULL    |                |
| status        | enum('PENDING','COMPLETED','REFUNDED') | YES  |     | PENDING |                |

Query results operations

Print
 Copy to clipboard
 Create view

Bookmark this SQL query

Label: 
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Bookmark this SQL query

Your SQL query has been executed successfully.

DESC Breed;

[\[ Edit inline \]](#)
[\[ Edit \]](#)
[\[ Create PHP code \]](#)

Extra options

| Field       | Type                           | Null | Key | Default | Extra          |
|-------------|--------------------------------|------|-----|---------|----------------|
| breedID     | int(10) unsigned               | NO   | PRI | NULL    | auto_increment |
| species     | varchar(50)                    | NO   | MUL | NULL    |                |
| breedName   | varchar(100)                   | NO   |     | NULL    |                |
| size        | enum('SMALL','MEDIUM','LARGE') | YES  |     | MEDIUM  |                |
| lifespan    | varchar(30)                    | YES  |     | NULL    |                |
| temperament | varchar(255)                   | YES  |     | NULL    |                |

Query results operations

Console

2 b)

Your SQL query has been executed successfully.

```
SELECT COUNT(*) FROM UserAccount;
```

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Extra options

COUNT(\*)

10

Query results operations



Print



Copy to clipboard



Export



Display chart



Create view



Bookmark this SQL query

Label:

☐ Let every user access this bookmark

Bookmark this SQL query



Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available. [?](#)

Your SQL query has been executed successfully.

```
SELECT COUNT(*) FROM Shelter;
```

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Extra options

COUNT(\*)

10

Your SQL query has been executed successfully.

```
SELECT COUNT(*) FROM Pet;
```

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Extra options

COUNT(\*)

10

Query results operations



Print



Copy to clipboard



Export



Display chart



Create view



Bookmark this SQL query

Label:



Let every user access this bookmark

Bookmark this SQL query



Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available. [?](#)

Your SQL query has been executed successfully.

```
SELECT COUNT(*) FROM Vaccine;
```

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Extra options

COUNT(\*)

10

Query results operations

Your SQL query has been executed successfully.

```
SELECT COUNT(*) FROM PetVaccine;
```

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Extra options

COUNT(\*)

10

Query results operations



Print



Copy to clipboard



Export



Display chart



Create view



Bookmark this SQL query

Label:



Let every user access this bookmark

Bookmark this SQL query



Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Your SQL query has been executed successfully.

```
SELECT COUNT(*) FROM Adoption;
```

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Extra options

COUNT(\*)

10

Your SQL query has been executed successfully.

`SELECT COUNT(*) FROM Appointment;`

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Extra options

COUNT(\*)

11

Query results operations



Print



Copy to clipboard



Export



Display chart



Create view



Bookmark this SQL query

Label:



Let every user access this bookmark

Bookmark this SQL query

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Your SQL query has been executed successfully.

`SELECT COUNT(*) FROM CareLog;`

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Extra options

COUNT(\*)

10

Your SQL query has been executed successfully.

```
SELECT COUNT(*) FROM Payment;
```

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Extra options

COUNT(\*)

10

Query results operations

[Print](#) [Copy to clipboard](#) [Export](#) [Display chart](#) [Create view](#)

[Bookmark this SQL query](#)

Label:  ☐ Let every user access this bookmark

[Bookmark this SQL query](#)

 Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available. [?](#)

Your SQL query has been executed successfully.

```
SELECT COUNT(*) FROM Breed;
```

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Extra options

COUNT(\*)

10

### 3) Sample Data:

Here, we have attached screenshots of the populated data in our tables. We have inserted more than 10 rows of data into all the tables in our database (as specified).

✓ Showing rows 0 - 9 (10 total, Query took 0.0005 seconds.)

SELECT \* FROM UserAccount;

☐ Profiling [ [Edit inline](#) ] [ [Edit](#) ] [ [Explain SQL](#) ] [ [Create PHP code](#) ] [ [Refresh](#) ]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

|                          |  | userID | name          | email              | phone    | role    |
|--------------------------|--|--------|---------------|--------------------|----------|---------|
| <input type="checkbox"/> |  |        |               |                    |          |         |
|                          |  | 1      | Alex Carter   | alex@example.com   | 555-1001 | ADOPTER |
| <input type="checkbox"/> |  |        |               |                    |          |         |
|                          |  | 2      | Bri Chen      | bri@example.com    | 555-1002 | ADOPTER |
| <input type="checkbox"/> |  |        |               |                    |          |         |
|                          |  | 3      | Chris Diaz    | chris@example.com  | 555-1003 | ADOPTER |
| <input type="checkbox"/> |  |        |               |                    |          |         |
|                          |  | 4      | Dana Fox      | dana@example.com   | 555-1004 | ADOPTER |
| <input type="checkbox"/> |  |        |               |                    |          |         |
|                          |  | 5      | Eli Gomez     | eli@example.com    | 555-1005 | ADOPTER |
| <input type="checkbox"/> |  |        |               |                    |          |         |
|                          |  | 6      | Sam Patel     | sam@example.com    | 555-2001 | STAFF   |
| <input type="checkbox"/> |  |        |               |                    |          |         |
|                          |  | 7      | Taylor Kim    | taylor@example.com | 555-2002 | STAFF   |
| <input type="checkbox"/> |  |        |               |                    |          |         |
|                          |  | 8      | Jordan Lee    | jlee@example.com   | 555-2003 | STAFF   |
| <input type="checkbox"/> |  |        |               |                    |          |         |
|                          |  | 9      | Dr. Priya Rao | prao@example.com   | 555-3001 | VET     |
| <input type="checkbox"/> |  |        |               |                    |          |         |
|                          |  | 10     | Dr. Omar Ali  | oali@example.com   | 555-3002 | VET     |

☐ Check all | With selected: Edit Copy Delete Export

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

✓ Showing rows 0 - 9 (10 total, Query took 0.0005 seconds.)

SELECT \* FROM Shelter;

☐ Profiling [ [Edit inline](#) ] [ [Edit](#) ] [ [Explain SQL](#) ] [ [Create PHP code](#) ] [ [Refresh](#) ]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

|                          |  | shelterID | name               | address              | phone    |
|--------------------------|--|-----------|--------------------|----------------------|----------|
| <input type="checkbox"/> |  |           |                    |                      |          |
|                          |  | 1         | Downtown Paws      | 101 Main St, City    | 555-4001 |
| <input type="checkbox"/> |  |           |                    |                      |          |
|                          |  | 2         | Uptown Tails       | 202 Elm St, City     | 555-4002 |
| <input type="checkbox"/> |  |           |                    |                      |          |
|                          |  | 3         | Harbor Haven       | 303 Bay Rd, City     | 555-4003 |
| <input type="checkbox"/> |  |           |                    |                      |          |
|                          |  | 4         | Riverside Rescue   | 404 River Ave, City  | 555-4004 |
| <input type="checkbox"/> |  |           |                    |                      |          |
|                          |  | 5         | Sunny Park Shelter | 505 Park Ln, City    | 555-4005 |
| <input type="checkbox"/> |  |           |                    |                      |          |
|                          |  | 6         | Brookside Shelter  | 606 Brook St, City   | 555-4006 |
| <input type="checkbox"/> |  |           |                    |                      |          |
|                          |  | 7         | Maple Grove        | 707 Maple Ave, City  | 555-4007 |
| <input type="checkbox"/> |  |           |                    |                      |          |
|                          |  | 8         | Pine Ridge         | 808 Pine Rd, City    | 555-4008 |
| <input type="checkbox"/> |  |           |                    |                      |          |
|                          |  | 9         | Cedar Corner       | 909 Cedar Blvd, City | 555-4009 |
| <input type="checkbox"/> |  |           |                    |                      |          |
|                          |  | 10        | Willow Wings       | 111 Willow Dr, City  | 555-4010 |

☐ Check all | With selected: Edit Copy Delete Export

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None



SELECT \* FROM PetVaccine;

☐ Profiling [ Edit inline ] [ Edit ] [ Explain SQL ] [ Create PHP code ] [ Refresh ]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

|                          |                  | petID | vaccineID | vaccineDate | lotNo   |
|--------------------------|------------------|-------|-----------|-------------|---------|
| <input type="checkbox"/> | Edit Copy Delete | 1     | 1         | 2025-09-02  | RAB-001 |
| <input type="checkbox"/> | Edit Copy Delete | 1     | 2         | 2025-09-03  | DHP-101 |
| <input type="checkbox"/> | Edit Copy Delete | 2     | 4         | 2025-09-06  | FVR-221 |
| <input type="checkbox"/> | Edit Copy Delete | 3     | 1         | 2025-09-11  | RAB-002 |
| <input type="checkbox"/> | Edit Copy Delete | 4     | 4         | 2025-09-13  | FVR-222 |
| <input type="checkbox"/> | Edit Copy Delete | 5     | 2         | 2025-09-16  | DHP-102 |
| <input type="checkbox"/> | Edit Copy Delete | 6     | 9         | 2025-09-19  | CAL-009 |
| <input type="checkbox"/> | Edit Copy Delete | 7     | 1         | 2025-09-20  | RAB-003 |
| <input type="checkbox"/> | Edit Copy Delete | 8     | 7         | 2025-09-21  | LYM-777 |
| <input type="checkbox"/> | Edit Copy Delete | 9     | 4         | 2025-09-23  | FVR-223 |

☐ Check all | With selected: Edit Copy Delete Export

Showing rows 0 - 9 (10 total, Query took 0.0006 seconds.)

SELECT \* FROM Adoption;

☐ Profiling [ Edit inline ] [ Edit ] [ Explain SQL ] [ Create PHP code ] [ Refresh ]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

|                          |                  | adoptionID | petID | adopterID | status    | applicationDate     | approvalDate        | finalizationDate    |
|--------------------------|------------------|------------|-------|-----------|-----------|---------------------|---------------------|---------------------|
| <input type="checkbox"/> | Edit Copy Delete | 1          | 1     | 1         | APPLIED   | 2025-10-01 00:00:00 | NULL                | NULL                |
| <input type="checkbox"/> | Edit Copy Delete | 2          | 2     | 2         | APPROVED  | 2025-10-01 00:00:00 | 2025-10-03 00:00:00 | NULL                |
| <input type="checkbox"/> | Edit Copy Delete | 3          | 3     | 3         | APPLIED   | 2025-10-01 00:00:00 | NULL                | NULL                |
| <input type="checkbox"/> | Edit Copy Delete | 4          | 4     | 4         | FINALIZED | 2025-10-02 00:00:00 | 2025-10-03 00:00:00 | 2025-10-05 00:00:00 |
| <input type="checkbox"/> | Edit Copy Delete | 5          | 5     | 5         | APPLIED   | 2025-10-02 00:00:00 | NULL                | NULL                |
| <input type="checkbox"/> | Edit Copy Delete | 6          | 6     | 6         | APPROVED  | 2025-10-03 00:00:00 | 2025-10-05 00:00:00 | NULL                |
| <input type="checkbox"/> | Edit Copy Delete | 7          | 7     | 7         | APPLIED   | 2025-10-03 00:00:00 | NULL                | NULL                |
| <input type="checkbox"/> | Edit Copy Delete | 8          | 8     | 8         | APPLIED   | 2025-10-03 00:00:00 | NULL                | NULL                |
| <input type="checkbox"/> | Edit Copy Delete | 9          | 9     | 9         | REJECTED  | 2025-10-04 00:00:00 | NULL                | NULL                |
| <input type="checkbox"/> | Edit Copy Delete | 10         | 10    | 10        | FINALIZED | 2025-10-04 00:00:00 | 2025-10-05 00:00:00 | 2025-10-06 00:00:00 |

☐ Check all | With selected: Edit Copy Delete Export

Showing rows 0 - 10 (11 total, Query took 0.0004 seconds.)

SELECT \* FROM Appointment;

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

|                          |                    | appointmentID | petID | adopterID | shelterID | appointmentTime        | appointmentType |
|--------------------------|--------------------|---------------|-------|-----------|-----------|------------------------|-----------------|
| <input type="checkbox"/> | Edit  Copy  Delete |               | 1     | 1         | 1         | 1 2025-10-01 10:00:00  | VISIT           |
| <input type="checkbox"/> | Edit  Copy  Delete |               | 2     | 2         | 2         | 2 2025-10-01 11:00:00  | MEET&GREET      |
| <input type="checkbox"/> | Edit  Copy  Delete |               | 3     | 3         | 3         | 3 2025-10-01 12:00:00  | VISIT           |
| <input type="checkbox"/> | Edit  Copy  Delete |               | 4     | 4         | 4         | 4 2025-10-02 10:00:00  | VISIT           |
| <input type="checkbox"/> | Edit  Copy  Delete |               | 5     | 5         | 5         | 5 2025-10-02 11:00:00  | MEET&GREET      |
| <input type="checkbox"/> | Edit  Copy  Delete |               | 6     | 6         | 1         | 6 2025-10-03 09:30:00  | VET             |
| <input type="checkbox"/> | Edit  Copy  Delete |               | 7     | 7         | 2         | 7 2025-10-03 10:15:00  | VISIT           |
| <input type="checkbox"/> | Edit  Copy  Delete |               | 8     | 8         | 3         | 8 2025-10-03 11:45:00  | VISIT           |
| <input type="checkbox"/> | Edit  Copy  Delete |               | 9     | 9         | 4         | 9 2025-10-04 14:00:00  | VISIT           |
| <input type="checkbox"/> | Edit  Copy  Delete |               | 10    | 10        | 5         | 10 2025-10-04 15:30:00 | MEET&GREET      |
| <input type="checkbox"/> | Edit  Copy  Delete |               | 11    | 1         | 1         | 1 2025-10-10 09:00:00  | VISIT           |

Bookmark this SQL query

Showing rows 0 - 9 (10 total, Query took 0.0006 seconds.)

SELECT \* FROM CareLog;

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

|                          |                    | careID | petID | staffID | careDate              | careType   | notes          |
|--------------------------|--------------------|--------|-------|---------|-----------------------|------------|----------------|
| <input type="checkbox"/> | Edit  Copy  Delete |        | 1     | 1       | 6 2025-09-02 08:00:00 | Feeding    | Kibble + water |
| <input type="checkbox"/> | Edit  Copy  Delete |        | 2     | 2       | 7 2025-09-06 08:00:00 | Cleaning   | Litter changed |
| <input type="checkbox"/> | Edit  Copy  Delete |        | 3     | 3       | 8 2025-09-11 08:00:00 | Exercise   | Short walk     |
| <input type="checkbox"/> | Edit  Copy  Delete |        | 4     | 4       | 6 2025-09-13 13:00:00 | Grooming   | Brushed fur    |
| <input type="checkbox"/> | Edit  Copy  Delete |        | 5     | 5       | 7 2025-09-16 09:00:00 | Feeding    | Wet food       |
| <input type="checkbox"/> | Edit  Copy  Delete |        | 6     | 6       | 8 2025-09-19 10:00:00 | Medication | Eye drops      |
| <input type="checkbox"/> | Edit  Copy  Delete |        | 7     | 7       | 6 2025-09-20 11:00:00 | Cleaning   | Hutch cleaned  |
| <input type="checkbox"/> | Edit  Copy  Delete |        | 8     | 8       | 7 2025-09-21 12:00:00 | Exercise   | Fetch          |
| <input type="checkbox"/> | Edit  Copy  Delete |        | 9     | 9       | 8 2025-09-23 08:30:00 | Feeding    | Dry food       |
| <input type="checkbox"/> | Edit  Copy  Delete |        | 10    | 10      | 6 2025-09-25 09:15:00 | Grooming   | Nail trim      |

Showing rows 0 - 9 (10 total, Query took 0.0022 seconds.)

SELECT \* FROM Payment;

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

|                          |                    | paymentID | adopterID | adoptionID | appointmentID | amount | paymentDate                | paymentMethod | status    |
|--------------------------|--------------------|-----------|-----------|------------|---------------|--------|----------------------------|---------------|-----------|
| <input type="checkbox"/> | Edit  Copy  Delete |           | 1         | 1          | 1             | NULL   | 25.00 2025-10-01 10:05:00  | CARD          | COMPLETED |
| <input type="checkbox"/> | Edit  Copy  Delete |           | 2         | 2          | 2             | NULL   | 25.00 2025-10-03 09:05:00  | CASH          | COMPLETED |
| <input type="checkbox"/> | Edit  Copy  Delete |           | 3         | 3          | 3             | NULL   | 25.00 2025-10-01 12:05:00  | CARD          | COMPLETED |
| <input type="checkbox"/> | Edit  Copy  Delete |           | 4         | 4          | 4             | NULL   | 150.00 2025-10-05 16:00:00 | CARD          | COMPLETED |
| <input type="checkbox"/> | Edit  Copy  Delete |           | 5         | 5          | 5             | NULL   | 25.00 2025-10-02 11:10:00  | CASH          | COMPLETED |
| <input type="checkbox"/> | Edit  Copy  Delete |           | 6         | 1          | NULL          | 6      | 40.00 2025-10-03 09:40:00  | CARD          | COMPLETED |
| <input type="checkbox"/> | Edit  Copy  Delete |           | 7         | 2          | NULL          | 7      | 25.00 2025-10-03 10:20:00  | CARD          | COMPLETED |
| <input type="checkbox"/> | Edit  Copy  Delete |           | 8         | 3          | NULL          | 8      | 25.00 2025-10-03 11:50:00  | CASH          | COMPLETED |
| <input type="checkbox"/> | Edit  Copy  Delete |           | 9         | 4          | NULL          | 9      | 25.00 2025-10-04 14:10:00  | CARD          | COMPLETED |
| <input type="checkbox"/> | Edit  Copy  Delete |           | 10        | 5          | 10            | NULL   | 150.00 2025-10-06 10:00:00 | CARD          | COMPLETED |

☐ Check all | With selected: Edit Copy Delete Export

Bookmark this SQL query

Showing rows 0 - 9 (10 total, Query took 0.0004 seconds.)

`SELECT * FROM Breed;`

☐ Profiling [ [Edit inline](#) ] [ [Edit](#) ] [ [Explain SQL](#) ] [ [Create PHP code](#) ] [ [Refresh](#) ]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

|                                           | breedID | species | breedName          | size   | lifespan  | temperament      |
|-------------------------------------------|---------|---------|--------------------|--------|-----------|------------------|
| <input type="checkbox"/> Edit Copy Delete | 1       | Dog     | Labrador Retriever | LARGE  | 10-12 yrs | Friendly, active |
| <input type="checkbox"/> Edit Copy Delete | 2       | Dog     | Beagle             | MEDIUM | 12-15 yrs | Curious, merry   |
| <input type="checkbox"/> Edit Copy Delete | 3       | Dog     | Poodle             | MEDIUM | 12-15 yrs | Smart, active    |
| <input type="checkbox"/> Edit Copy Delete | 4       | Cat     | Domestic Shorthair | MEDIUM | 12-16 yrs | Affectionate     |
| <input type="checkbox"/> Edit Copy Delete | 5       | Cat     | Siamese            | SMALL  | 10-12 yrs | Vocal, social    |
| <input type="checkbox"/> Edit Copy Delete | 6       | Cat     | Maine Coon         | LARGE  | 10-13 yrs | Gentle           |
| <input type="checkbox"/> Edit Copy Delete | 7       | Rabbit  | Holland Lop        | SMALL  | 7-10 yrs  | Calm             |
| <input type="checkbox"/> Edit Copy Delete | 8       | Rabbit  | Rex                | MEDIUM | 7-10 yrs  | Docile           |
| <input type="checkbox"/> Edit Copy Delete | 9       | Dog     | German Shepherd    | LARGE  | 9-13 yrs  | Confident        |
| <input type="checkbox"/> Edit Copy Delete | 10      | Cat     | Bengal             | MEDIUM | 12-16 yrs | Energetic        |

#### 4) SQL commands used:

##### SECTION 4A: NON-ADVANCED SQL COMMANDS

###### 1. CREATE TABLE Commands

**Table 1:** UserAccount

```
CREATE TABLE UserAccount (
  userID INT UNSIGNED AUTO_INCREMENT,
  name VARCHAR(100) NOT NULL,
  email VARCHAR(50) NOT NULL UNIQUE,
  phone VARCHAR(30),
  role ENUM('ADOPTER', 'STAFF', 'VET', 'ADMIN') NOT NULL,
  PRIMARY KEY (userID)
);
```

**Table 2:** Shelter

```
CREATE TABLE Shelter (
  shelterID INT UNSIGNED AUTO_INCREMENT,
  name VARCHAR(100) NOT NULL,
  address VARCHAR(100) NOT NULL,
  phone VARCHAR(30),
  PRIMARY KEY (shelterID)
```

);

**Table 3:** Pet

```
CREATE TABLE Pet (  
    petID INT UNSIGNED AUTO_INCREMENT,  
    shelterID INT UNSIGNED NOT NULL,  
    name VARCHAR(100),  
    species VARCHAR(50),  
    breed VARCHAR(50),  
    sex ENUM('M', 'F', 'UNKNOWN') NOT NULL DEFAULT 'UNKNOWN',  
    dob DATE,  
    status ENUM('AVAILABLE', 'HOLD', 'ADOPTED') NOT NULL DEFAULT 'AVAILABLE',  
    intakeDate DATE NOT NULL,  
    PRIMARY KEY (petID),  
    CONSTRAINT FK_PET_SHELTER FOREIGN KEY (shelterID) REFERENCES  
    Shelter(shelterID) ON UPDATE CASCADE ON DELETE RESTRICT  
);
```

**Table 4:** Vaccine

```
CREATE TABLE Vaccine (  
    vaccineID INT UNSIGNED AUTO_INCREMENT,  
    name VARCHAR(120) NOT NULL UNIQUE,  
    PRIMARY KEY (vaccineID)  
);
```

**Table 5:** PetVaccine

```
CREATE TABLE PetVaccine (  
    petID INT UNSIGNED NOT NULL,  
    vaccineID INT UNSIGNED NOT NULL,  
    vaccineDate DATE NOT NULL,  
    lotNo VARCHAR(64) NOT NULL,  
    PRIMARY KEY (petID, vaccineID, vaccineDate),  
    CONSTRAINT FK_PV_PET FOREIGN KEY (petID) REFERENCES Pet(petID) ON  
    UPDATE CASCADE ON DELETE CASCADE,  
    CONSTRAINT FK_PV_VACCINE FOREIGN KEY (vaccineID) REFERENCES  
    Vaccine(vaccineID) ON UPDATE CASCADE ON DELETE RESTRICT  
);
```

**Table 6:** Adoption

```
CREATE TABLE Adoption (  
    adoptionID INT UNSIGNED AUTO_INCREMENT,  
    petID INT UNSIGNED NOT NULL,  
    adopterID INT UNSIGNED NOT NULL,
```

```

    status ENUM('APPLIED', 'APPROVED', 'FINALIZED', 'REJECTED', 'CANCELED') NOT
    NULL,
    applicationDate DATETIME,
    approvalDate DATETIME,
    finalizationDate DATETIME,
    PRIMARY KEY (adoptionID),
    CONSTRAINT FK_ADOPT_PET FOREIGN KEY (petID) REFERENCES Pet(petID) ON
    UPDATE CASCADE ON DELETE RESTRICT,
    CONSTRAINT FK_ADOPT_ADOPTER FOREIGN KEY (adopterID) REFERENCES
    UserAccount(userID) ON UPDATE CASCADE ON DELETE RESTRICT,
    CONSTRAINT UQ_PET_FINALIZED UNIQUE (petID, finalizationDate)
);

```

**Table 7:** Appointment

```

CREATE TABLE Appointment (
    appointmentID INT UNSIGNED AUTO_INCREMENT,
    petID INT UNSIGNED NOT NULL,
    adopterID INT UNSIGNED NOT NULL,
    shelterID INT UNSIGNED NOT NULL,
    appointmentTime DATETIME NOT NULL,
    appointmentType ENUM('VISIT', 'MEET&GREET', 'VET') NOT NULL,
    PRIMARY KEY (appointmentID),
    CONSTRAINT FK_APPT_PET FOREIGN KEY (petID) REFERENCES Pet(petID) ON
    UPDATE CASCADE ON DELETE RESTRICT,
    CONSTRAINT FK_APPT_ADOPTER FOREIGN KEY (adopterID) REFERENCES
    UserAccount(userID) ON UPDATE CASCADE ON DELETE RESTRICT,
    CONSTRAINT FK_APPT_SHELTER FOREIGN KEY (shelterID) REFERENCES
    Shelter(shelterID) ON UPDATE CASCADE ON DELETE RESTRICT
);

```

**Table 8:** CareLog

```

CREATE TABLE CareLog (
    careID INT UNSIGNED AUTO_INCREMENT,
    petID INT UNSIGNED NOT NULL,
    staffID INT UNSIGNED NOT NULL,
    careDate DATETIME NOT NULL,
    careType VARCHAR(80) NOT NULL,
    notes TEXT,
    PRIMARY KEY (careID),
    CONSTRAINT FK_CL_PET FOREIGN KEY (petID) REFERENCES Pet(petID) ON
    UPDATE CASCADE ON DELETE CASCADE,
    CONSTRAINT FK_CL_STAFF FOREIGN KEY (staffID) REFERENCES
    UserAccount(userID) ON UPDATE CASCADE ON DELETE RESTRICT

```

);

**Table 9:** Payment

```
CREATE TABLE Payment (  
  paymentID INT UNSIGNED AUTO_INCREMENT,  
  adopterID INT UNSIGNED NOT NULL,  
  adoptionID INT UNSIGNED,  
  appointmentID INT UNSIGNED,  
  amount DECIMAL(10,2) NOT NULL,  
  paymentDate DATETIME NOT NULL,  
  paymentMethod ENUM('CASH', 'CARD') NOT NULL,  
  status ENUM('PENDING', 'COMPLETED', 'REFUNDED') DEFAULT 'PENDING',  
  PRIMARY KEY (paymentID),  
  CONSTRAINT FK_PAY_ADOPTER FOREIGN KEY (adopterID) REFERENCES  
  UserAccount(userID) ON UPDATE CASCADE ON DELETE RESTRICT,  
  CONSTRAINT FK_PAY_ADOPT FOREIGN KEY (adoptionID) REFERENCES  
  Adoption(adoptionID) ON UPDATE CASCADE ON DELETE SET NULL,  
  CONSTRAINT FK_PAY_APPT FOREIGN KEY (appointmentID) REFERENCES  
  Appointment(appointmentID) ON UPDATE CASCADE ON DELETE SET NULL  
);
```

**Table 10:** Breed

```
CREATE TABLE Breed (  
  breedID INT UNSIGNED AUTO_INCREMENT,  
  species VARCHAR(50) NOT NULL,  
  breedName VARCHAR(100) NOT NULL,  
  size ENUM('SMALL', 'MEDIUM', 'LARGE') DEFAULT 'MEDIUM',  
  lifespan VARCHAR(30),  
  temperament VARCHAR(255),  
  PRIMARY KEY (breedID),  
  UNIQUE (species, breedName)  
);
```

## 2. RETRIEVE (SELECT) Commands

### **Query 1:** Browse Available Pets with Filters

Purpose: Display available pets with optional filtering by species, breed, and sex with pagination

```
SELECT p.*, s.name AS shelterName  
FROM Pet p JOIN Shelter s ON s.shelterID=p.shelterID  
WHERE (p.status = 'AVAILABLE')  
AND (COALESCE(?species, p.species) = p.species)
```

```
AND (COALESCE(?breed, p.breed) = p.breed)
AND (COALESCE(?sex, p.sex) = p.sex)
ORDER BY p.intakeDate DESC
LIMIT ?limit OFFSET ?offset;
```

**Query 2:** Pet Details with Age and Shelter

Purpose: Display complete pet details including calculated age for individual pet view

```
SELECT p.*, fn_pet_age_years(p.petID) AS ageYears, s.name AS shelterName
FROM Pet p JOIN Shelter s ON s.shelterID=p.shelterID
WHERE p.petID = ?petID;
```

**Query 3:** Adopter Dashboard - Adoption Pipeline

Purpose: Show all adoption applications for a specific adopter with status tracking

```
SELECT * FROM vw_adoption_pipeline
WHERE adopterID = ?userID
ORDER BY applicationDate DESC;
```

**Query 4:** Vaccination History for Pet

Purpose: Display complete vaccination records for medical tracking

```
SELECT v.name, pv.vaccineDate, pv.lotNo
FROM PetVaccine pv JOIN Vaccine v ON v.vaccineID=pv.vaccineID
WHERE pv.petID=?petID
ORDER BY pv.vaccineDate DESC;
```

**Query 5:** Available Pets View

Purpose: Simplified view showing all available pets with shelter information

```
SELECT p.petID, p.name, p.species, p.breed, p.sex, p.dob, p.intakeDate, s.name AS
shelterName
FROM Pet p JOIN Shelter s ON p.shelterID = s.shelterID
WHERE p.status = 'AVAILABLE';
```

**Query 6:** All Staff Members

Purpose: Display all staff users for scheduling and management

```
SELECT userID, name, email, phone
FROM UserAccount
WHERE role = 'STAFF';
```

**Query 7:** Adoption Pipeline Status Summary

Purpose: View all adoptions with applicant names and pet names

```
SELECT a.adoptionID, a.petID, p.name AS petName, a.adopterID, u.name AS
adopterName, a.status,
a.applicationDate, a.approvalDate, a.finalizationDate
FROM Adoption a
```

```
JOIN Pet p ON p.petID = a.petID
JOIN UserAccount u ON u.userID = a.adopterID;
```

**Query 8: Care Log History**

Purpose: Track all care activities for a specific pet

```
SELECT cl.careID, cl.careDate, cl.careType, cl.notes, ua.name AS staffName
FROM CareLog cl
JOIN UserAccount ua ON cl.staffID = ua.userID
WHERE cl.petID = ?petID
ORDER BY cl.careDate DESC;
```

**Query 9: Payment Records for Adopter**

Purpose: Display payment history for a specific adopter

```
SELECT p.paymentID, p.amount, p.paymentDate, p.paymentMethod, p.status,
       CASE WHEN p.adoptionID IS NOT NULL THEN 'Adoption' ELSE 'Appointment'
END AS paymentType
FROM Payment p
WHERE p.adopterID = ?adopterID
ORDER BY p.paymentDate DESC;
```

**Query 10: Pets by Shelter**

Purpose: Show all pets at a specific shelter

```
SELECT p.petID, p.name, p.species, p.breed, p.status
FROM Pet p
WHERE p.shelterID = ?shelterID
ORDER BY p.intakeDate DESC;
```

### **3. ADD (INSERT) Commands**

**Insert 1: Create Adoption Application**

Purpose: Submit new adoption application for a pet

```
INSERT INTO Adoption(petID, adopterID, status, applicationDate)
VALUES (?petID, ?adopterID, 'APPLIED', NOW());
```

**Insert 2: Add Staff Care Log Entry**

Purpose: Log daily care activities for pets (feeding, exercise, grooming, medication)

```
INSERT INTO CareLog(petID, staffID, careDate, careType, notes)
VALUES (?petID, ?staffID, NOW(), ?careType, ?notes);
```

**Insert 3: Record Payment Transaction**

Purpose: Record payment from adopter for adoption fees or appointment services

```
INSERT INTO Payment(adopterID, adoptionID, appointmentID, amount, paymentDate,
paymentMethod, status)
VALUES (?adopterID, ?adoptionID, ?appointmentID, ?amount, NOW(), ?method,
'COMPLETED');
```

**Insert 4:** Add New User Account

Purpose: Register new user (adopter, staff, veterinarian)

```
INSERT INTO UserAccount(name, email, phone, role)
VALUES (?name, ?email, ?phone, ?role);
```

**Insert 5:** Register New Pet

Purpose: Add newly rescued or accepted pet to the system

```
INSERT INTO Pet(shelterID, name, species, breed, sex, dob, status, intakeDate)
VALUES (?shelterID, ?name, ?species, ?breed, ?sex, ?dob, 'AVAILABLE', NOW());
```

**Insert 6:** Record Vaccination

Purpose: Document pet vaccination with lot number for medical records

```
INSERT INTO PetVaccine(petID, vaccineID, vaccineDate, lotNo)
VALUES (?petID, ?vaccineID, ?vaccineDate, ?lotNo);
```

#### 4. UPDATE Commands

**Update 1:** Approve Adoption Application

Purpose: Staff approves an adoption application, moving it from APPLIED to APPROVED status

```
UPDATE Adoption
SET status='APPROVED', approvalDate=NOW()
WHERE adoptionID=?adoptionID AND status='APPLIED';
```

**Update 2:** Finalize Adoption

Purpose: Complete the adoption process, marking pet as ADOPTED

```
UPDATE Adoption
SET status='FINALIZED', finalizationDate=NOW()
WHERE adoptionID=?adoptionID AND status='APPROVED';
```

**Update 3:** Reject Adoption Application

Purpose: Staff rejects an adoption application

```
UPDATE Adoption
SET status='REJECTED'
WHERE adoptionID=?adoptionID AND status='APPLIED';
```

**Update 4: Update Pet Status**

Purpose: Change pet availability status (AVAILABLE, HOLD, ADOPTED)

UPDATE Pet

SET status=?newStatus

WHERE petID=?petID;

**Update 5: Update User Information**

Purpose: Modify user profile details

UPDATE UserAccount

SET name=?name, email=?email, phone=?phone

WHERE userID=?userID;

**Update 6: Update Payment Status**

Purpose: Change payment status (PENDING, COMPLETED, REFUNDED)

UPDATE Payment

SET status=?newStatus

WHERE paymentID=?paymentID;

**5. DELETE Commands****Delete 1: Remove Pet Record**

Purpose: Delete pet from system (use with caution - cascades to related records)

DELETE FROM Pet

WHERE petID = ?petID;

**Delete 2: Cancel Adoption Application**

Purpose: Delete or cancel an adoption application

DELETE FROM Adoption

WHERE adoptionID = ?adoptionID;

**Delete 3: Remove Care Log Entry**

Purpose: Delete erroneous care log entry

DELETE FROM CareLog

WHERE careID = ?careID;

**Delete 4: Remove Payment Record**

Purpose: Delete payment transaction (for refunds or corrections)

DELETE FROM Payment

WHERE paymentID = ?paymentID;

**Delete 5: Remove Appointment**

Purpose: Cancel scheduled appointment

```
DELETE FROM Appointment
WHERE appointmentID = ?appointmentID;
```

## **SECTION 4B: ADVANCED PL/SQL COMMANDS**

### **1. TRIGGER:** Role Validation for Adoption

Command:

```
DELIMITER $$
CREATE TRIGGER trg_adoption_role_check
BEFORE INSERT ON Adoption
FOR EACH ROW
BEGIN
    DECLARE v_role ENUM('ADOPTER','STAFF','VET','ADMIN');
    SELECT role INTO v_role FROM UserAccount WHERE userID = NEW.adopterID;
    IF v_role IS NULL OR v_role <> 'ADOPTER' THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Only users with role=ADOPTER can adopt.';
    END IF;
END $$
DELIMITER ;
```

#### Purpose:

Enforces business logic at the database level. Prevents non-adopter users (STAFF, VET, ADMIN) from creating adoption applications. Automatically validates role before any adoption record is inserted. Maintains data integrity by rejecting invalid adoption requests.

#### When Used:

Triggered automatically whenever an adoption application is submitted. Runs BEFORE INSERT on the Adoption table. Returns error if user role is not ADOPTER.

### **2. FUNCTION:** Calculate Pet Age

Command:

```
DELIMITER $$
CREATE FUNCTION fn_pet_age_years(p_petID INT)
RETURNS DECIMAL(4,1)
```

```

DETERMINISTIC
BEGIN
  DECLARE v_dob DATE;
  DECLARE v_age DECIMAL(4,1);
  SELECT dob INTO v_dob FROM Pet WHERE petID = p_petID;
  IF v_dob IS NULL THEN RETURN NULL; END IF;
  SET v_age = TIMESTAMPDIFF(MONTH, v_dob, CURDATE())/12.0;
  RETURN ROUND(v_age,1);
END $$
DELIMITER ;

```

Purpose:

Calculates pet age dynamically based on date of birth. Avoids storing static age that becomes outdated. Converts age to years with one decimal place for readability. Handles NULL dates gracefully.

When Used:

Called in SELECT queries to display pet age: SELECT fn\_pet\_age\_years(petID) AS ageYears

Used in pet detail views and browsing interfaces. Automatically updates as time passes without manual updates.

**Returns:**

Decimal number (e.g., 3.2 years) rounded to one decimal place. NULL if pet has no date of birth.

### 3. **STORED PROCEDURE:** Safe Appointment Scheduling

Command:

```

DELIMITER $$
CREATE PROCEDURE sp_schedule_appointment(
  IN p_petID INT,
  IN p_adopterID INT,
  IN p_shelterID INT,
  IN p_time DATETIME,
  IN p_type ENUM('VISIT','MEET&GREET','VET')
)
BEGIN
  IF NOT EXISTS (SELECT 1 FROM Pet WHERE petID = p_petID) THEN
    SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT='Invalid petID';
  END IF;

```

```

IF NOT EXISTS (SELECT 1 FROM UserAccount WHERE userID = p_adopterID AND
role='ADOPTER') THEN
    SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT='Invalid adopterID or role';
END IF;
IF NOT EXISTS (SELECT 1 FROM Shelter WHERE shelterID = p_shelterID) THEN
    SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT='Invalid shelterID';
END IF;
IF EXISTS (SELECT 1 FROM Appointment WHERE petID=p_petID AND
appointmentTime = p_time) THEN
    SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT='Pet already has an appointment
at that time';
END IF;
INSERT INTO Appointment(petID, adopterID, shelterID, appointmentTime,
appointmentType)
VALUES (p_petID, p_adopterID, p_shelterID, p_time, p_type);
END $$
DELIMITER ;

```

**Purpose:**

Encapsulates complex business logic for appointment scheduling. Validates all foreign keys and business rules before insertion. Prevents double-booking of pets at the same time. Ensures data consistency and integrity.

Validations Performed:

1. Pet exists in database
2. User is registered as an ADOPTER
3. Shelter exists in database
4. No existing appointment for this pet at the exact same time

**Error Handling:**

Returns specific error messages for each validation failure. Prevents invalid data from being inserted. Maintains referential integrity.

**4. STORED PROCEDURE:** Seed Database with Test Data

Command:

```

DELIMITER $$
CREATE PROCEDURE sp_seed_data()
BEGIN
    SET FOREIGN_KEY_CHECKS=0;
    TRUNCATE TABLE Payment;

```

```
TRUNCATE TABLE CareLog;
TRUNCATE TABLE Appointment;
TRUNCATE TABLE PetVaccine;
TRUNCATE TABLE Adoption;
TRUNCATE TABLE Vaccine;
TRUNCATE TABLE Pet;
TRUNCATE TABLE Breed;
TRUNCATE TABLE Shelter;
TRUNCATE TABLE UserAccount;
SET FOREIGN_KEY_CHECKS=1;
```

```
INSERT INTO UserAccount(name,email,phone,role) VALUES
('Alex Carter','alex@example.com','555-1001','ADOPTER'),
('Bri Chen','bri@example.com','555-1002','ADOPTER'),
('Chris Diaz','chris@example.com','555-1003','ADOPTER'),
('Dana Fox','dana@example.com','555-1004','ADOPTER'),
('Eli Gomez','eli@example.com','555-1005','ADOPTER'),
('Sam Patel','sam@example.com','555-2001','STAFF'),
('Taylor Kim','taylor@example.com','555-2002','STAFF'),
('Jordan Lee','jlee@example.com','555-2003','STAFF'),
('Dr. Priya Rao','prao@example.com','555-3001','VET'),
('Dr. Omar Ali','oali@example.com','555-3002','VET');
```

```
INSERT INTO Shelter(name,address,phone) VALUES
('Downtown Paws','101 Main St, City','555-4001'),
('Uptown Tails','202 Elm St, City','555-4002'),
('Harbor Haven','303 Bay Rd, City','555-4003'),
('Riverside Rescue','404 River Ave, City','555-4004'),
('Sunny Park Shelter','505 Park Ln, City','555-4005'),
('Brookside Shelter','606 Brook St, City','555-4006'),
('Maple Grove','707 Maple Ave, City','555-4007'),
('Pine Ridge','808 Pine Rd, City','555-4008'),
('Cedar Corner','909 Cedar Blvd, City','555-4009'),
('Willow Wings','111 Willow Dr, City','555-4010');
```

```
INSERT INTO Breed(species,breedName,size,lifespan,temperament) VALUES
('Dog','Labrador Retriever','LARGE','10-12 yrs','Friendly, active'),
('Dog','Beagle','MEDIUM','12-15 yrs','Curious, merry'),
('Dog','Poodle','MEDIUM','12-15 yrs','Smart, active'),
('Cat','Domestic Shorthair','MEDIUM','12-16 yrs','Affectionate'),
('Cat','Siamese','SMALL','10-12 yrs','Vocal, social'),
('Cat','Maine Coon','LARGE','10-13 yrs','Gentle'),
('Rabbit','Holland Lop','SMALL','7-10 yrs','Calm');
```

```
('Rabbit','Rex','MEDIUM','7-10 yrs','Docile'),
('Dog','German Shepherd','LARGE','9-13 yrs','Confident'),
('Cat','Bengal','MEDIUM','12-16 yrs','Energetic');
```

```
INSERT INTO Vaccine(name) VALUES
('Rabies'),('DHPP'),('Bordetella'),('FVRCP'),('FeLV'),
('Leptospirosis'),('Lyme'),('Parvovirus Booster'),('Calicivirus'),('Panleukopenia');
```

```
INSERT INTO Pet(shelterID,name,species,breed,sex,dob,status,intakeDate) VALUES
(1,'Buddy','Dog','Labrador Retriever','M','2021-04-15','AVAILABLE','2025-09-01'),
(2,'Luna','Cat','Domestic Shorthair','F','2023-02-20','AVAILABLE','2025-09-05'),
(3,'Max','Dog','Beagle','M','2022-06-10','HOLD','2025-09-10'),
(4,'Milo','Cat','Siamese','M','2024-01-05','AVAILABLE','2025-09-12'),
(5,'Bella','Dog','Poodle','F','2020-08-08','AVAILABLE','2025-09-15'),
(6,'Cleo','Cat','Maine Coon','F','2021-11-11','AVAILABLE','2025-09-18'),
(7,'Nibbles','Rabbit','Holland Lop','F','2024-04-01','AVAILABLE','2025-09-19'),
(8,'Scout','Dog','German Shepherd','M','2022-09-09','AVAILABLE','2025-09-20'),
(9,'Ziggy','Cat','Bengal','M','2023-05-22','AVAILABLE','2025-09-22'),
(10,'Pumpkin','Rabbit','Rex','F','2024-06-30','AVAILABLE','2025-09-25');
```

```
INSERT INTO PetVaccine(petID,vaccineID,vaccineDate,lotNo) VALUES
(1,1,'2025-09-02','RAB-001'),
(1,2,'2025-09-03','DHP-101'),
(2,4,'2025-09-06','FVR-221'),
(3,1,'2025-09-11','RAB-002'),
(4,4,'2025-09-13','FVR-222'),
(5,2,'2025-09-16','DHP-102'),
(6,9,'2025-09-19','CAL-009'),
(7,1,'2025-09-20','RAB-003'),
(8,7,'2025-09-21','LYM-777'),
(9,4,'2025-09-23','FVR-223');
```

```
INSERT INTO
Appointment(petID,adopterID,shelterID,appointmentTime,appointmentType) VALUES
(1,1,1,'2025-10-01 10:00:00','VISIT'),
(2,2,2,'2025-10-01 11:00:00','MEET&GREET'),
(3,3,3,'2025-10-01 12:00:00','VISIT'),
(4,4,4,'2025-10-02 10:00:00','VISIT'),
(5,5,5,'2025-10-02 11:00:00','MEET&GREET'),
(6,1,6,'2025-10-03 09:30:00','VET'),
(7,2,7,'2025-10-03 10:15:00','VISIT'),
(8,3,8,'2025-10-03 11:45:00','VISIT'),
(9,4,9,'2025-10-04 14:00:00','VISIT');
```

```
(10,5,10,'2025-10-04 15:30:00','MEET&GREET');
```

```
INSERT INTO
```

```
Adoption(petID,adopterID,status,applicationDate,approvalDate,finalizationDate) VALUES  
(1,1,'APPLIED','2025-10-01',NULL,NULL),  
(2,2,'APPROVED','2025-10-01','2025-10-03',NULL),  
(3,3,'APPLIED','2025-10-01',NULL,NULL),  
(4,4,'FINALIZED','2025-10-02','2025-10-03','2025-10-05'),  
(5,5,'APPLIED','2025-10-02',NULL,NULL),  
(6,1,'APPROVED','2025-10-03','2025-10-05',NULL),  
(7,2,'APPLIED','2025-10-03',NULL,NULL),  
(8,3,'APPLIED','2025-10-03',NULL,NULL),  
(9,4,'REJECTED','2025-10-04',NULL,NULL),  
(10,5,'FINALIZED','2025-10-04','2025-10-05','2025-10-06');
```

```
INSERT INTO CareLog(petID,staffID,careDate,careType,notes) VALUES
```

```
(1,6,'2025-09-02 08:00:00','Feeding','Kibble + water'),  
(2,7,'2025-09-06 08:00:00','Cleaning','Litter changed'),  
(3,8,'2025-09-11 08:00:00','Exercise','Short walk'),  
(4,6,'2025-09-13 13:00:00','Grooming','Brushed fur'),  
(5,7,'2025-09-16 09:00:00','Feeding','Wet food'),  
(6,8,'2025-09-19 10:00:00','Medication','Eye drops'),  
(7,6,'2025-09-20 11:00:00','Cleaning','Hutch cleaned'),  
(8,7,'2025-09-21 12:00:00','Exercise','Fetch'),  
(9,8,'2025-09-23 08:30:00','Feeding','Dry food'),  
(10,6,'2025-09-25 09:15:00','Grooming','Nail trim');
```

```
INSERT INTO
```

```
Payment(adopterID,adoptionID,appointmentID,amount,paymentDate,paymentMethod,status) VALUES
```

```
(1,1,NULL,25.00,'2025-10-01 10:05:00','CARD','COMPLETED'),  
(2,2,NULL,25.00,'2025-10-03 09:05:00','CASH','COMPLETED'),  
(3,3,NULL,25.00,'2025-10-01 12:05:00','CARD','COMPLETED'),  
(4,4,NULL,150.00,'2025-10-05 16:00:00','CARD','COMPLETED'),  
(5,5,NULL,25.00,'2025-10-02 11:10:00','CASH','COMPLETED'),  
(1,NULL,6,40.00,'2025-10-03 09:40:00','CARD','COMPLETED'),  
(2,NULL,7,25.00,'2025-10-03 10:20:00','CARD','COMPLETED'),  
(3,NULL,8,25.00,'2025-10-03 11:50:00','CASH','COMPLETED'),  
(4,NULL,9,25.00,'2025-10-04 14:10:00','CARD','COMPLETED'),  
(5,10,NULL,150.00,'2025-10-06 10:00:00','CARD','COMPLETED');
```

```
END $$
```

```
DELIMITER ;
```

```
]
```